
EE/CprE/SE 491 WEEKLY REPORT 5

Mar 26, 2025 - Apr 1, 2025

Group number: 16

Project title: Outdoor Roomba

Client &/Advisor: Prof. Diane Rover

Team Members/Role:

Brodie M Bates

Daniel Vergara Pinilla

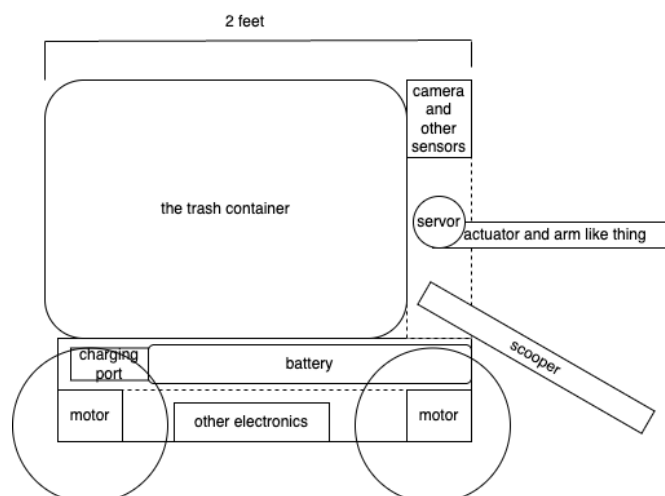
Mitchell D Owen

Om Shukla

Sudipta Halder

○ Weekly Summary

This week we decided to make a big design change. Previously we had a truck-like design. But it had many flaws. It used one motor preside so when turning it would turn like a tank which made it ineffective when making a turn and had a higher possibility of hitting nearby object. But the biggest problem was its size. It was too big. It took longer to build and there was no place to store it over the summer, autonomous navigation was much harder to account for. Therefore we decided to make the size a 2 feet weight and 2 feet long as you can see in the figure below a rough sketch of the side-view of our new design. All the sensors and electronics are the same except we made it less longer.



○ **Past week accomplishments**

- Sudipta Halder: work on redesign on the robot, made it smaller so we can finish building it early. Looked through all designs of the robot in sample project from Webots and checked if it was within our budget and fit our requirements, in the non fit our requirements. Looked at sensors for the roomba and motors.
- Brodie Bates: Work on redesign and sizing, research sensors and parts to use, and added to bill of materials.
- Daniel Pinilla: Work on the Ai direction model for robot movement
- Om Shukla: Learning Webots to help model robot functionality. Also started learning about geofencing.
- Mitchell Owen: Researching parts and looking at new Webots to integrate

○ **Pending issues**

- Brodie Bates: Need to finalize a design and decide on what to do with the server to start.
- Daniel Pinilla: finish movement management code
- Om Shukla: Finish learning Webots and make progress learning about geofencing and how it can be implemented for our project.
- Mitchell Owen: Purchases parts for the robot.

○ **Individual contributions**

<u>NAME</u>	<u>Individual Contributions</u>	<u>Hours this week</u>	<u>HOURS cumulative</u>
Brodie Bates	Research sensors, redraw robot,	7	29
Om Shukla	Researched more on Webots and geofencing	6	23
Sudipta Halder	Redesign the robot. Looked at 72 Webot sample robots. Research on the sensors	7	32
Daniel Mauricio Vergara	AI object detection and movement direction	7	32
Mitchell Owen	Researching Webots	3	25

○ **Plans for the upcoming week**

- Sudipta Halder: Complete the simulation bit on Webot. Get user interface implemented with Om in the Webot simulation. So we can get ready for the slides
- Brodie Bates: CAD new idea for design, research sensors, find good placement for sensors, prototype design.
- Om Shukla: Learn more about Webots and model robot functionality like moving and object detection and object marking in a geofence. Learn more about geofencing and how it works.
- Mitchell Owen: See if we can order some items from the BOM to start building.
- Daniel Mauricio Vergara: 3D modeling and working on websocket code for the robot

○ **Summary of weekly advisor meeting**

This week for the advisor meeting we were told to sketch out our system including server side and what's on the robot. Also to talk to ETG and see what parts we could get from previous senior design projects that they have. Lastly, we asked questions about CAN protocol, and our upcoming presentation on user testing.